System audit, 9 October 2026

What the system proves, what is wrong in it, what was corrected during the audit, and the order in which the rest should be done

9 October 2026

Prepared by Dupleix Institute (Edward Mazibuko, Director, Data Analytics) for MAIIC (Dr Thomson Kumwenda, CFO) and the Dupleix engagement file

Addendum of 14 October 2026 at the end gives the status of every finding against the code at abe140d plus the uncommitted working tree; dated status notes are added where a 'Closed' label overstated. Original scope: Scope is the MAICC-IFRS9 system at commit f5b08fc on branch eir_revenue_recognition, audited against specification v4 of 7 October 2026, IFRS 9, the RBM Credit Risk Management for DFIs Directive 2018, and ordinary standards of security, data integrity and operability. Confidential to MAIIC and Dupleix.

1. Verdict

The system does what the After Build Report of 8 October says it does in the sense that every step named there runs from one command on a wiped database and produces the figures quoted. The audit went behind those figures, and the honest summary is this: **the data foundation is sound and proven to the cent; the impairment engine is not yet on an IFRS 9 basis; and the legacy modules the platform was forked from expose the whole loan book and its governed inputs without a login.** None of the three can be left as it is before MAIIC relies on a number from the system or puts it on a network.

In order of consequence:

1. **The ECL is a 12-month ECL for every stage.** Stage 2 loans (8.35bn of exposure, 36 percent of the book at August 2026) are provided at the 12-month PD, not the lifetime PD; the lifetime PD the engine computes is written to a column nothing reads. The allowance is also undiscounted in the bootstrap. The 8.94bn quoted in the After Build Report understates a lifetime, discounted ECL by an amount that cannot be stated without running the correct engine, but a Stage 2 loan with 36 months to run and a 10 percent annual PD is provided at 10 percent where IFRS 9 requires 27.
2. **Stage 3 interest is recognised on the gross amount**, not the net carrying amount, because the revenue engine reads an allowance column (`expected_loss_provision`) that the governed build never writes. The "net basis" setting is in force and the code path exists; it has no allowance to net. The test that covers it seeds the column by hand.
3. **Write-offs (type 300) and the Nascomex equity entries (type 401) are treated as cash received** by the loan-book build and the EIR roll-forward, against the specification's own list at section 3.3 (22.3m and 535.7m in the committed pack).
4. **180 web routes carry no authentication**, including the loan-book import, the collateral register, the transition-matrix and LGD writes into the loan book, the ECL recalculation and a one-click regression approval. One of them interpolates request input into a raw SQL statement. All of these are modules of the product the platform was forked from, not work of this engagement, but they are on the server MAIIC will run.
5. **The RBM classification bands built on 8 October were wrong for medium- and long-term facilities** (they put a 31-to-90-day facility in special mention at 5 percent where the Gazette says standard at 0, and used 360/720 where the Gazette says 365/746). This was Dupleix's error, found by this audit and corrected the same day at commit `f5b08fc`; the test now asserts every boundary in the Gazette.

The rest of this report gives the evidence, the full list of findings by severity, what was corrected during the audit, and the order in which the remainder should be done.

2. How the audit was done

Everything below rests on things that were run or read, not on the After Build Report's own claims.

Proof	Result
Clean install: `eir:bootstrap --fresh --force-wipe --with-client-inputs --build --run-engines --verify` on a wiped throwaway database	Completed in 183 seconds; 9 of 9 golden checks pass (ledger interest 5,293,988,207.06; method B equals the stored report on 2,361 of 2,361 account-months; the December 2025 GL ties incl. the two accepted FInES exceptions; the 4215/4216 year-end legs).
Every IFRS 9 report rendered for August 2026 (`ifrs9:smoke-reports`, now all 30 methods)	All OK after the audit's own fixes (two reports had thrown a division error on a sector with no exposure; see section 6).
Full test suite	See section 7 for the counts; the suite cannot complete in one process under XAMPP's 300-second execution limit, which is itself a finding.
Dependency audit (`composer audit`, `npm audit --omit=dev`)	Composer: two medium advisories (symfony/routing CVE-2026-48784 and CVE-2026-45065), one low (symfony/polyfill-intl-idn), two abandoned packages (beyondcode/laravel-websockets, box/spout). npm: 33 advisories (1 critical, 19 high) led by TinyMCE XSS in the help editor and a yaml stack overflow.
Route inventory (`route:list` with controller middleware merged) and a read of every `$this->middleware` in the controllers	180 of 724 web routes have no `auth` middleware.
Four independent code reviews, each read-only with file-and-line evidence: specification conformance (sections 3 to 16, every decision, the 48-setting catalogue), financial-engine correctness, security and access control, data integrity and operability	The reviews' findings were re-read against the code before inclusion; the Critical items in section 4 were each re-verified by hand.
The Gazette itself (`docs/regulatory`, pages 683 and 685)	Read in full for the classification bands and rates.
The bootstrapped database queried directly for coverage	135 loans in the August 2026 book; 144 contracts in the EIR master; 31 schedules generated and 12 EIRs locked (section 5).

3. What the system proves (and should keep)

- **The ledger is the record and the build reproduces it.** Method B derives every account-month's carrying amount from the landed ledger and equals E-Banker's own stored report to the cent on 2,361 of 2,361 account-months; the loan GLs tie to the December 2025 trial balance to the cent, with the two FInES exceptions carried as accepted exceptions rather than hidden.
- **Landing, gates and versioning.** Packs land with hashes, counts, dates, key uniqueness and query-id checks; rows are append-only and versioned with their pack; a failed gate quarantines the load.
- **Maker-checker is enforced server-side** on every path built in this engagement (loan-book build, governance settings, scenario sets, regression fits, compliance sign-off); the bootstrap's own approvals carry a label that is visibly not a person's.
- **The EIR solver** is sound: Newton with bisection fallback, bounded, residual re-checked; locked EIRs cannot be recomputed; reopening archives and voids downstream rows.
- **Staging** reads days past due from the oldest overdue instalment under the governed basis, applies the directive's Stage 3 thresholds by tenor class (91 days short-term, 181 otherwise), allocates receipts to instalments in due-date order for the four-instalment trigger, and writes the post-qualitative stage the ECL and FLI read.

- **Units are consistent where it matters most**: the facility utilisation rate is stored as a fraction and read as one by every EAD reader; PD is stored in percent in the matrix and divided into a fraction by both PD writers; Stage 3 PD is forced to 1 in every engine.
- **The forward-looking chain** fails closed (observations, sign, R-squared, p-value), declines with reasons, and the post-FLI PD is bounded and weighted by validated weights.
- **The suite shell and theme**: every one of 120 captured screens renders in dark mode (measured, not assumed) and the light theme is unchanged.
- **Governance of the feed token**: the API route compares the bearer token in constant time and refuses an empty configured token.

4. Findings

Closure, 9 October 2026 (same day, after the audit's first issue). Step 8 followed the same evening on four branches merged at `2903d66` (gates and the trial balances through the door; the overlay register, the lock tie and the set editor; the cure period, PD on the measured stage, the legacy regression retired, macro permissions and two screens; the take-on recompute, the as-at conventions and exports, the sign-test and precedence settings wired, the zip guard). Open now: H12 and M18 (D30), H13's missing ties (data MAIIC confirms), and the hygiene of step 9. **Status 14 October 2026:** more is open than this sentence says. Against the code at HEAD abe140d, 18 findings are only partly closed (C1, C2, H3, H4, H7, H9, H11, H13, H14, M2, M8, M9, M12, M15, M17, L2, L4, L5), H12 and M18 wait on D30, and L1, L3 and L6 are not done; see the addendum at the end of this report. The "doable" steps of section 9 (1 to 7) were done and committed on `eir_revenue_recognition`: `d58a8b0` (C7, C8), `ba117ff` (C1, C2, C3, H7, M3), `7d339a6` (C5, C6, M17), `bf616a7` (H1, M15), `d062a53` (C9, H3, M8, M9), `4a31816` (H4, H11, H9), `af95179` (H2, H8, H5, H6), with `4a98dca` (H10) and `f5b08fc` (C4) from earlier in the day. Each finding below carries **Closed** with its commit where that is so; the figures in section 8 are restated at the end. The open findings are those that need MAIIC (H12, H13's missing ties) or the remaining Medium and Low items.

Severity: **Critical** = a wrong number in a financial statement or a regulatory return, or an unauthenticated path to financial data; **High** = wrong under a realistic condition, or a control that can be bypassed; **Medium** = a fragile or undocumented assumption, or a gap against the specification that does not yet move a number; **Low** = cosmetic or latent. Each finding names the file so an engineer can go straight to it.

4.1 Critical

#	Finding	Where	Effect and example	Fix
C1	**Closed** `ba117ff`**. Status 14 October 2026:** partly closed. The lifetime PD for Stage 2 is built (`ExpectedCreditLossController.php:409-421`). Since `216aee2` the month-end and the bootstrap run the ECL discounted (`FliMonthEndCommand.php:85`, `SegmentEcl.php:42`, `Bootstrap.php:347,417`), and the time-phased engine no longer overwrites `ecl_value` (`TimePhasedEclService.php:35-46`). But the discounted allowance sits beside the booked figure in `ecl_value_discounted`: the FS note and the reports still read the undiscounted `ecl_value` (`Ifrs9NoteService.php:192,220`). Stage 2 allowance is a 12-month ECL; `lifetime_pd` is written and never read; the bootstrap books an undiscounted ECL	`ExpectedCreditLossController.php:375-383` (ecl = pd × lgd × EAD with the 12-month PD for every stage); `RecalculateEcl.php:24` defaults `--discounting=undiscounted` and `Bootstrap.php:315` accepts the default; `PdEngineService.php:71` writes `lifetime_pd`	Stage 2, 12m PD 10%, 36 months remaining, LGD 0.5, EAD 1,000,000: booked 50,000; lifetime $1-0.9^3$ = 27.1% gives 135,500 before discounting. Every report, the FS note and the RBM return read `ecl_value`	Stage 2 uses `lifetime_pd`; the bootstrap runs the time-phased engine (`--discounting=discounted`) once C2 and C3 are fixed; the FS note then carries lifetime, discounted figures. **Status 14 October 2026:** the note carries a lifetime figure but not a discounted one, and its text still describes the measure as EAD x PD x LGD (`Ifrs9NoteService.php:488`). To close: the note and the reports read `ecl_value_discounted` where the run was discounted, and the note states the basis
C2	**Closed** `ba117ff`**. Status 14 October 2026:** partly closed. Every main reader takes months, but the legacy Internal Grading screen still reads months as years (`InternalGradingController.php:411`, `tenor_years = (int) remaining_tenor`). `remaining_tenor` is months from the governed build and years from the legacy importer; readers disagree	Writes months: `LoanBookBuildService.php:549`. Writes years: `ProcessLoanImportJob.php`, `LoanBooksImport.php`. Reads months: `PdEngineService.php:70`. Reads years: `TransitionMatrixController.php:523,546` (lifetime PD = 1−(1−pd)^tenor), `TimePhasedEclService.php:169` (×12), `EclDiscountingService.php:81`	Under build B a 24-month Stage 2 loan: the Monthly Probability screen's lifetime PD is 1−0.9^24 = 92% against the engine's 19%; the discounting service discounts over 24 "years"	One unit (months), stated on the column, and every reader converted; a test per reader
C3	**Closed** `ba117ff`**.** Stage 3 ECL in the time-phased engine is discounted from the wrong date and the recovery plan is netted before discounting	`TimePhasedEclService.php:69,105,114-115`	EAD 100,000, LGD 0.6, 24-month schedule, EIR 10%: 49,587 instead of 60,000; the covering test asserts the understated range	Loss at the reporting date, or EAD less the present value of each recovery from its own date; the LGD fed in is the undiscounted cohort LGD (`LgdEngineService.php:70`) and should be discounted
C4	RBM classification bands wrong for medium- and long-term facilities (built 8 October)	`RbmReturnService.php:36`, `Ifrs9ReportsController.php` (RBM constant and `rbmClassCase`), `tests/Feature/Rbm/RbmReturnServiceTest.php`	A 36-month facility 60 days overdue on 10m: minimum provision 500,000 instead of 0; 361-365 days long-term Doubtful (50%) not Substandard (20%); 721-746 Loss not Doubtful	**Corrected at** `f5b08fc` to the Gazette's section 10 (short-term 30/90/180/365; medium and long 90/180/365/746); the test asserts every boundary on both terms; workbook 4 F3 records both corrections

#	Finding	Where	Effect and example	Fix
C5	**Closed** `7d339a6`**.** The revenue roll-forward deducts scheduled instalments as cash after the borrower's last receipt	`EirRevenueService.php:171-178`: "covered" is the contract's own first and last transaction; after the last receipt every month falls back to the schedule	Loan 1,000 at 1% a month, 100 a month schedule, last receipt March: by December the engine's gross is about 100 while the ledger shows about 1,100. This is the Stage 3 population	The window is the feed's coverage (`LandingZoneReader::lastLedgerDate()`), not the contract's; a month inside coverage with no receipt is zero cash
C6	**Closed** `7d339a6`**.** Stage 3 "net carrying amount" interest never takes effect	`EirRevenueService.php:70` reads `loan_books.expected_loss_provision`; the governed build's column list (`LoanBookBuildService.php:45-53`) does not write it; only the legacy `LoanBookController.php:331` and test data do. The bootstrap also runs revenue before the ECL (`Bootstrap.php:226-258` then `:315`)	Gross 1,000, allowance 600, 1% a month: recognised 10, should be 4, on every Stage 3 loan (8.26bn of exposure)	Read `ecl_value` (the allowance the engine writes); run the ECL for the prior period before the revenue step, or carry the opening allowance on the roll-forward row
C7	**Closed** `d58a8b0`**.** 180 web routes have no authentication	`routes/web.php` modules: `loss-given-default` (22 routes), `transition-matrix` (19), `loan_application` (18), `collateral` (13), `credit-loss-data` (11), `transition-matrix-cummulative` (10), `lgd-calculations` (9), `regression` (9), `file` (9), `expected-credit-loss` (5), `groups`, `scenarios`, `scenario-profiles`, `macro-forecast-weighted`, `macro-statistics` (legacy), `transition-profile`, `api/*`	`POST /transition-matrix/{id}/update-loan-book` rewrites the PDs that feed the ECL; `POST /expected-credit-loss/calculations` recomputes the allowance; `GET /api/collateral-registers` returns the whole collateral register as JSON with CORS open to any origin; `PATCH /regression/{model}/approve` approves a model with no user and no log. None needs a cookie	Wrap every legacy module in `auth` plus a permission; delete the dead `routes/api.php` routes (two name controller methods that do not exist); this is a day's work and the first thing to do before the server is reachable from the MAIIC network
C8	**Closed** `d58a8b0`**.** Unauthenticated SQL interpolation	`TransitionProfileOptionController.php:66-96` (`POST /transition-profile/categories/reorder`): `categories.*.id` validated with `exists:` then interpolated into `DB::statement("... WHEN {$id} THEN ...")`. MySQL's numeric coercion lets `"5 OR 1=1"` pass an `exists` check against id 5	Arbitrary SQL in the CASE expression from an unauthenticated request	Bind the ids; cast to int; authenticate the route (C7)

#	Finding	Where	Effect and example	Fix
C9	**Closed** `d062a53`**. Status 14 October 2026:** closed for the bootstrap. One residual remains: a null approver also skips the different-person check (`FliRouteService.php:71`), so `fli:apply --approve` run without a user lets one person approve. The bootstrap approves a regression fit under its own label and computes the ECL on it	`Bootstrap.php:298-309` proposes, approves and applies the best fit; spec 6.10.1 step 6 says the overlay is zero and the post-FLI PD marked "no adjustment: bootstrap" until two people approve a fit; the impairment workbook's own finding F2 states the spec's rule	The 8.94bn ECL (and the 10.24bn weighted sensitivity) in the After Build Report rests on a fit no MAIIC person approved; `FliRouteService.php:52` lets any non-null label bypass the different-person check	The bootstrap applies the route with the overlay at zero and marks the PD; fit approval is a screen action by two people; the label bypass is limited to the bootstrap's constant

4.2 High

#	Finding	Where	Effect	Fix
H1	**Closed** `bf616a7`**. Status 14 October 2026:** closed. Since `220a5a6`, `WriteOffClassifier` also treats as write-offs the batches that debit GL 1064 and the ACCOUNT CLOSURE credits on status-H loans, and the register `loan_write_offs` is under maker-checker (`WriteOffTest`, 5 tests). The amounts written off still do not reach the IFRS 7 reconciliation, which hard-codes nil (`EclReconciliationService.php:383,416`; `LoanBookReconciliationService.php:64`). Write-offs (300) and the Nascomex equity entries (401) are classed as cash receipts	`LoanBookBuildService.php:42` (`TYPE_RECEIPT` includes 401 and 300); `ContractInputsBuildService.php:176-180`	22.3m of write-offs and 535.7m of equity entries in the committed pack flow into `repayments` and the EIR roll-forward's `cash_received`. Spec 3.3 lists 300 as write-off and 400/401 as equity. The method B tie to the stored report still passes because the stored report makes the same mistake; the tie proves reproduction, not correctness	A write-off reduces gross carrying through the allowance, not cash; 401 is excluded from loan cash; re-prove the ties with the exceptions stated
H2	**Closed** `af95179`**.** Tranche facilities: later drawdowns never enter the EIR cash-flow vector	`EirContractInputService.php:81-82` uses `drawn_amount` at origination and the version-1 schedule; `DisbursementService::laterDrawdownFlows` is unused; `EirReadinessService.php:30,44-49`	Approved 100m, 60m drawn at origination and 40m six months later: the solver sees 60m out and 100m plus interest in; the EIR solves far above the contractual rate. Spec v4 counts 58 tranched facilities	Negative (disbursement) flows at their dates from P3_12; a test on a two-tranche facility
H3	**Closed** `d062a53`**. Status 14 October 2026:** partly closed. The time-phased engine uses the governed set written on each loan (`TimePhasedEclService.php:80,164`). But the legacy synthetic set (BASE 60, UPSIDE 20, DOWNSIDE 20) is still seeded APPROVED (`Bootstrap.php:106`, `EclScenarioAssumptionSeeder`) and weights every loan that carries no governed set. Scenario weighting is applied twice over two different sets	`FliRouteService.php:127-136` writes the probability-weighted PD across the governed set; `TimePhasedEclService.php:82-85` then weights again by `ecl_scenario_assumptions` (a seeded "controlled synthetic" set, BASE 60 / UPSIDE 20 / DOWNSIDE 20), which `Bootstrap.php:105` seeds as approved	Double-weighting whenever the time-phased engine runs; and two different per-scenario answers for the same period (the sensitivity's typed multipliers against the route's regression adjustments: 10.24bn against 8.94bn in the After Build Report)	One governed set; the ECL per scenario, weighted once, and the sensitivity reads the same per-scenario ECLs (spec 15.5)

#	Finding	Where	Effect	Fix
H4	**Closed** 4a31816**. Status 14 October 2026:** partly closed. `ReportingPeriodLock` is checked by staging, PD, LGD, the FLI route, segment ECL, revenue and the write-off register. It is not checked by `MegaFarmEclService`, by `EirCalculationService` (which clears discounted figures on loan_books) or by the legacy transition-matrix, LGD, collateral and grading writers. The reporting-period lock is honoured only by the loan-book build	`LoanBookBuildService.php:616` is the sole reader of `reporting_period_locks`; `EirRevenueService::run(recalculate: true)` supersedes rows in a locked period; ECL, staging, PD, LGD and FLI write freely	A locked month can be restated by any engine; `EirAsAtService.php:66-69` claims the opposite	One lock check in a shared helper, called by every writer; a test per writer
H5	**Closed** af95179**.** Reports and the FS note split by the pre-qualitative stage; the ECL was measured on the post-qualitative stage	`Ifrs9ReportsController.php:137,161,268,301,333,367,390,419,628`; `ExpectedCreditLossController.php:420`; `StressTestService.php:25-26,44`	A loan moved to Stage 3 by the four-instalment trigger (post-only) appears in "Stage 1" with PD 1.0 in "Closing ECL by Stage" and the impairment charge by stage	Every reader splits by `ifrs9_stage_post_qualitative`
H6	**Closed** af95179**.** The FS note's "gross carrying amount" is the EAD	`Ifrs9ReportsController.php:628-633` uses carrying + commitments × utilisation as gross carrying; NPL ratios at `:573` and `:1240` are on EAD	Carrying 60m, commitments 40m, utilisation 0.6: "gross carrying" 84m in the note	Gross carrying = carrying amount; EAD is its own column
H7	**Closed** ba117ff**. Status 14 October 2026:** partly closed. Write-off as loss, cure and recovery kept exclusive, and discounting are built (`LgdEngineService.php`; since 216aee2 a written-off loan left at nil is no longer counted as a cure). Collateral is not used: one portfolio LGD applies to every loan. LGD cohort workout counts a written-off loan as fully recovered and double-counts a cured loan that also paid down; recoveries undiscounted; no collateral	`LgdEngineService.php:38,49-56,70`	Cohort of two 500s, one written off (absent from the end book): recovery 50 percent, LGD 0.5; the true recovery on the absent loan is 0	Absent-from-book is a write-off unless the ledger shows settlement; cure and recovery are exclusive; discount at the EIR; collateral from the register
H8	**Closed** af95179**.** 12-month PD from a short window is not annualised	`PdEngineService.php:36-38,65-67`	With books from October 2025 run for January 2026, the "12-month PD" is a three-month default rate	Annualise 1−(1−p)^(12/months) and record the window

#	Finding	Where	Effect	Fix
H9	**Closed** 4a31816**. Status 14 October 2026:** partly closed. The administrator override is a governed setting (GovernanceService.php:462). Schedule approval still has no different-person check (ScheduleWorkflowService.php:305-319), and the setting schedule_approval_control is read by no code. The admin role bypasses maker-checker on governance approvals and EIR locks	EirGovernanceController.php:67-70 (adminOverride() = has role admin), used at ten call sites; EirCalculationService.php:108-111	An administrator can propose and approve a governed setting alone, and lock an EIR they calculated	The override is a governed setting with its own approval, or removed; schedule approval (ScheduleWorkflowService.php:275) should also require a different person from the editor
H10	Governance defaults were effective from 1 January 2025; 63 of 144 contracts originated before 2024	GovernanceSettingsSeeder.php (EFFECTIVE_FROM); GovernanceService::inForce() resolves at the contract's origination date	16 contracts on the clean install could not generate a schedule because "no approved value is in force for day_count" at origination	**Fixed (9 October, after the audit's first issue):** defaults seeded from inception; a migration backdates the untouched defaults on installed databases (49 rows on the demo copy, audit-logged); a test proves every catalogue setting resolves on MAIIC's earliest origination date (4 March 2020). On the clean install schedules rose from 31 to 47 and locked EIRs from 12 to 27; golden checks unchanged at 9 of 9
H11	**Closed** 4a31816**. Status 14 October 2026:** partly closed, as section 9 step 6 already says. The thresholds are resolved at the period end with no silent default (StagingService), but staging_thresholds has no propose-and-approve path (Settings shows them read only), and the legacy importer's StagingClassifier still resolves at today's date with a 31/181 fallback (StagingClassifier.php:84,95; StagingThreshold.php:43). Staging thresholds are resolved at today's date, not the period end, and sit outside maker-checker	StagingThreshold.php:36-46 uses now(); StagingService.php:113 silently defaults to [31,181]	A re-run of a past period uses today's thresholds, against D21; a missing row is not an error	Resolve at the period end; thresholds proposed and approved like every other setting; no silent default

#	Finding	Where	Effect	Fix
H12	Mega Farm ECL misreads decision D30. **Status 14 October 2026:** open, waiting on D30. Since `216aee2` the programme is kept out of the main PD, LGD and ECL: segment ECL leaves out the programme portfolio and refuses a mixed book (`SegmentEcl.php:69-72,121-126`; `MegaFarmProgrammeSeparationTest`), so a portfolio run no longer restores the loan to 100 percent. The programme ECL is still EAD x PD x LGD x share (`MegaFarmEclService.php:166`), and the share falls back to 1.0 when the setting names none (`:93`)	`MegaFarmEclService.php:58,117-125`: `ecl_value = EAD × PD × LGD × 5%`	The loan allowance is booked at 5 percent of the programme loss. D30: the provision is the fund's; MAIIC's 5 percent is a share of interest, written down separately. Neither the fund-side provision nor the receivable write-down is produced; a later portfolio ECL run restores the loan to 100 percent (`ExpectedCreditLossController.php:375` has no GL exclusion)	Awaits Dr Thom's confirmation of D30 as stated in the After Build Report; until then the Mega Farm figure must not be quoted
H13	The audit's own `--verify` covers 9 checks against 13 section-9 ties and 6 golden numbers. **Status 14 October 2026:** section 9 of the specification now has 16 rows (13 on 9 October). `--verify` asserts five kinds of check (`BaselineService.php:27-81`): the ledger interest total, method B against the stored report, the 4215 leg, the 4216 leg, and the December 2025 loan GLs against the trial balance (one row per GL, hence the nine rows). It also reports, but does not test, the Mega Farm difference. The other 11 rows of section 9 are stated only, and there are still no ECL or revenue-shift golden numbers	`BaselineService.php`	No 2024 year-end GL tie, no monthly GL tie, no interest-income YTD tie, no 28-posting count, no feed re-pull test, no collateral or take-on counts; the ECL golden numbers await the first approved run	Add the missing ties as the data for each is confirmed; the After Build Report's "9 checks, 0 FAIL" is accurate about what runs and should say so

#	Finding	Where	Effect	Fix
H14	The full test suite could not run in one process	Eleven legacy controllers call `ini_set('max_execution_time', 300)` for their own long requests (`ExpectedCreditLossController.php:287`, `LoanBookController.php:562`, `CollateralController.php:324`, the LGD controllers); once a test touches one, every test that follows shares a five-minute clock and the suite dies with "Maximum execution time exceeded" (three attempts, each from a different test)	A CI or release check that runs "the tests" silently runs part of them; the After Build Report's "308 passed" was a directory run, not the suite	**Fixed:** `tests/TestCase::setUp` resets the limit to unbounded before every test; the controllers should raise the limit only when it is lower (not lower it from unlimited). **Status 14 October 2026:** partly closed. The suite half is done (`tests/TestCase.php:19`). Since `216aee2` the bootstrap, `ifrs9:segment-ecl` and `ifrs9:smoke-reports` reset the limit to unlimited after each call (`Bootstrap.php:348,495`, `SegmentEcl.php:82`, `SmokeIfrs9Reports.php:39`). The ECL controller lowers it only for a web request, but that change is uncommitted. Eight other controllers still lower an unlimited limit to 300

4.3 Medium

#	Finding	Where
M1	**Closed** 2903d66**.** Trial balances bypass the landing zone (imported directly, no hash, no load row); GL_02 is landed but not read by the reconciliation	`Bootstrap.php:142-147`; `TrialBalanceImportService.php`; `EirGlReconciliationService.php`
M2	**Closed** 2903d66**. Status 14 October 2026:** partly closed. The account, numeric and query-version gates fail a pack, but the ISO-date, balance-history and month-end-row gates only warn (`PackGates.php:144,228,273`). The fee GL gate was built in `216aee2` as a report on `eir:build-gl-fees` (`GlFeeBuildService.php:286-312`), not as a gate at the door. Gates missing at the door: no account-in-master check, no balance-history-to-ledger tie, no month-end-row-per-account check, no numeric parse, query version recorded but not validated; quarantine keeps no rows; dates accepted in whatever format the manifest declares (the committed pack is m/d/yyyy, against D23)	`PackLandingService.php:65-129,145-148,288-299`
M3	**Closed** 2903d66**.** The manual-overlay register of spec 14.6 does not exist; the "Manual overlay" route reads the latest legacy `fli_adj` row	`FliRouteService.php:106-111`
M4	**Closed** 2903d66**.** The overlay-lock rule of 15.7 is an empty `if` block; `lock()` checks nothing	`ScenarioSetService.php:91-93,154-162`
M5	**Closed** 2903d66**.** The Scenario Sets screen cannot set weights, add scenarios or edit shocks; only the fixed first set can be seeded. The decision the spec reserves for the CFO has no screen	`ScenarioSetController.php`; `Pages/Governance/ScenarioSets.vue`
M6	**Closed** 2903d66**.** Two regression chains run in parallel: the new univariate path under maker-checker, and the legacy screen (`RegressionController.php:227-233`) with one-click approval, no test and no second person, still in the menu as "Regression Analysis"	
M7	**Closed** 2903d66**.** Take-on "recompute from origination" writes a label and nothing else: no EIR solve from the workbook schedule, no `TAKEON_WORKBOOK` schedule	`TakeonLandingService.php:179-228`
M8	**Closed. Status 14 October 2026:** partly closed. The bootstrap no longer reviews fee classifications. It still locks every solved EIR by administrator override (`Bootstrap.php:314`, `lock(..., true)`); the run note now names this as an override under the bootstrap label. The bootstrap gives Governance-Centre-type approvals it is told never to give: fee classifications reviewed under the label, EIR locks by administrator override; and, inconsistently, leaves the 30 seeded fee rules unapproved, so 19 contracts stall at the fee gate and EIR coverage on the clean install is 12 of 135 loans	`Bootstrap.php:182-189,242`; `FeeRuleMatcher.php:72-77`
M9	**Closed. Status 14 October 2026:** partly closed. The EAD half is fixed. `coalesce(pd_prefli, pd_value, 12m_pd)` remains (`ScenarioSetService.php:557`, `OverlayService.php:226`), and `12m_pd` is still a decimal(8,2) column written by more than one writer (`SeedReportFields.php:81`). Sensitivity reads `coalesce(pd_prefli, pd_value, 12m_pd)` where `12m_pd` is percent from one writer and a fraction from another; and uses the stored `ead` column, which the build fills at 100 percent utilisation unlike every other EAD reader	`ScenarioSetService.php:252-257`; `LoanBookBuildService.php:555`; `SeedReportFields.php:81`

#	Finding	Where
M10	**Closed** 2903d66**.** PD is assigned by the pre-qualitative stage and the matrix is pre-to-pre; the instalment trigger is outside the PD's default definition; loans that leave the book count as "Paid" in the denominator	`PdEngineService.php:67,69`; `MaiicTransitionProfileSeeder.php:22`
M11	**Closed** 2903d66**.** No cure or probation period: stage is recomputed from current DPD only; the prior stage is selected and never read	`StagingService.php:53-71`
M12	**Closed** 2903d66**. Status 14 October 2026:** partly closed. The catalogue now holds 61 settings, and 20 of them are read by no engine code. They are plr_mid_period, reset_trigger, rate_source_precedence, margin_basis, cash_source, manual_policy_handling, modification_threshold, counter_reset_handling, contractual_record, expected_cashflow_basis, history_before_dec_2025, trueup_gl_account, auditor_export_format, keyman_insurance_treatment, nascomex_preference_shares, staging_rebuttal, rate_change_classification, schedule_approval_control, historic_materiality_assessment and fee_reclass_journal. `partly_drawn_interest_basis` has been read since `216aee2`. The second store `governed_parameters` remains (`GovernedValues.php`). Fourteen catalogue settings are read by nothing but the Governance Centre and the trace (`fli_expected_sign_test`, `macro_source_precedence`, `ebanker_feed_route` and others); the FLI thresholds live in a second store (`governed_parameters`) with different seed defaults that a direct write would bypass	`GovernanceService.php`; `GovernedValues.php:38-64`; `PollFeedFolder.php:37-41`
M13	**Closed** 2903d66**. Status 14 October 2026:** closed as asked. Review item R12 of 12 October found a separate defect in the part-month branch: the contractual accrual is returned as `eir_period_to_date` (`EirAsAtService.php:123-127,188`). It is not fixed. As-at date: CSV only (spec: Excel and PDF); part-month accrues on the EIR compounded rather than the section-3 convention; take-on basis not read	`EirAsAtService.php:71-75`; `EirAsAtController.php:76`
M14	**Closed** 2903d66**.** Zip extraction from the feed API and the poller relies on PHP's internal path sanitisation with no explicit check against `..` entries	`routes/web.php:1319`; `PollFeedFolder.php`
M15	**Closed. Status 14 October 2026:** partly closed. Only `COLLECTION_TYPES` is fixed (`EirRevenueService.php:29`). Reversal pairing is not built: a receipt reversal only nets the month's cash (`EirLedgerCashService.php:85`). The over-sanction flag is not built. The `RATE_BASIS` mapping is still in the legacy importer (`ContractMasterImport.php:71-75`). `COLLECTION_TYPES` still contains 'Interest' (spec 7.4 says corrected); reversal pairing (7.1) not found; over-sanction flag (7.3) not found; `RATE_BASIS` mapping still in the legacy Excel importer	`EirRevenueService.php:29`; `ContractMasterImport.php:71-75`
M16	**Closed** 2903d66**.** Help chapters still carry the old group names; Correlation Finder and Auditor Pack are CLI-only with no screen; `macro.view` / `macro.manage` permissions do not exist (macro commit uses `eir.govern`)	help seeders `07a`, `08`; `config/menu.php`; `MacroStatisticsController.php:26`

#	Finding	Where
M17	**Closed. Status 14 October 2026:** partly closed. Since `216aee2` the bootstrap runs, for every period, the PD and LGD by segment, the FLI route, the discounted ECL and the Mega Farm programme (`Bootstrap.php:330-361`), as well as staging and revenue. The scenario-set seed, the correlation and fit, the sensitivity, the stress test and the report smoke still run for the last period only (`Bootstrap.php:379-451`). Engine chain in the bootstrap runs PD, LGD, FLI, ECL, stress and reports for the last period only, against "every period from the first full month"	`Bootstrap.php:263-367`
M18	**Status 14 October 2026:** open, waiting on D30. Since `216aee2` the programme's LGD is measured on its own cohort, a scalar that cannot be measured is declined rather than typed in (`MegaFarmEclService.php:117-120,228-244`), and the programme is a segment of its own. The interest split, the receivable, the fund liability and the report are not built. Section 16 beyond staging and PD is not built: the 15 percent interest split, the maize repayment and buyer receivable, the fund liability roll-forward, the Mega Farms report, seasonal segmentation; LGD borrowed from MAIIC's own book; scalar fallback hard-coded from the statements	`MegaFarmEclService.php:150-151,168-179`

4.4 Low and hygiene

#	Finding	Where
L1	The repository carries a 715-file `corporate/` folder (a credit-scoring model with `.exe`, `.rar`, raw InnoDB `.ibd` and SQL dumps) and a medical ICD-10 XML under `storage/app`, both from the product's initial commit, both deployed to the client's server	`corporate/`; `storage/app/icd10cm_tabular_2021.xml`
L2	Other-client strings outside `docs/reference`: FDH in method keys and worked examples the user sees (`AdjustmentMethods.php`, `TransmissionMethodCatalogue.php:181-183`, `GovernedValues.php:47,54`, `Guardrail.php:18`, `StressTestingController.php:14-15`); ZNBS in seeded support-ticket text (`TicketSeeder.php:234,253`); three committed session transcripts under `docs/sessions` naming other clients throughout; build scripts under `docs/build-files` with other-client paths. The compliance PDF footer was the worst instance and is fixed at `f5b08fc`	
L3	Composer advisories (symfony/routing ×2 medium, polyfill-intl-idn low), two abandoned packages; npm 33 advisories (TinyMCE XSS in the help editor, yaml)	`composer.lock`, `package-lock.json`
L4	`.env.example` and config: cache prefix was shared across the production, demo and bootstrap env files (fixed 8 October: keyed by `APP_ENV`); `SESSION_SECURE_COOKIE` unset by default; dead CSRF exclusion `file/upload`	`config/cache.php`; `config/session.php:171`; `VerifyCsrfToken.php:14`
L5	`ScheduleWorkflowService.php:76` treats a stated rate above 1 as a percent, so a stated "1" (1 percent) becomes 100 percent; `CalculateEirService.php:88` 30/360 is 30E/360; `lifetime_pd` and `12m_pd` are decimal(8,2) so an 8-dp write is truncated; `StagingService.php:156` casts `TRANSAMT` without stripping commas	
L6	Legacy test suites (92 "medical" tests and five FLI page tests) fail on `master` before this engagement and remain in the tree; test classes build their own sqlite schemas by hand, so the real migrations are exercised only by the bootstrap	`tests/`

5. Coverage of the EIR chain on a clean install

The After Build Report quotes "31 schedules, 31 EIRs solved, 12 locked". The audit traced the other 113 contracts and the 19 unlocked EIRs on the freshly bootstrapped database:

Reason the schedule was not generated	Contracts	Whose it is
A moratorium is stated but its type is not, on the contract or its scheme	51	MAIIC (a decision on the default moratorium type, or the scheme data)
Drawn amount not positive (with other reasons)	39	Data: undisbursed or held contracts; the readiness rule is right to refuse them
No approved value for `day_count` at the origination date	16, now 0	Dupleix (finding H10, fixed the same day; the 16 now generate)

Reason the schedule was not generated	Contracts	Whose it is
Part-drawn and neither contract nor scheme says whether interest is on the drawn amount	5	MAIIC (`partly_drawn_interest_basis` is a catalogue setting nothing reads yet; M12). **Status 14 October 2026:** the setting has been read since `216aee2`, after E-Banker's own postings (`ScheduleWorkflowService.php:375`); the value is still MAIIC's to approve
Contractual rate missing or invalid	2	Data
EIR solved but not locked: fee lines not independently reviewed	**19, now 20**	The 30 seeded fee rules are unapproved by design (Dr Thom's rulebook approval is owed); see M8 for the bootstrap's inconsistency

So on the clean install as first audited, the revenue roll-forward, the as-at service and the GL reconciliation covered 12 of 135 loans; after the H10 fix they cover 26 of 135 (27 locked EIRs), and the ECL covers all 135. The next two steps are MAIIC's: the moratorium type (51 contracts) and the fee rulebook (20 EIRs). These facts are now in the After Build Report's section 3 in those words.

6. Corrected during the audit

Each is committed with its evidence.

Commit	What
after `e176d1d`	Governance defaults from inception (H10): seeder, backdating migration, test; schedules 31 to 47 and locked EIRs 12 to 27 on the clean install
`f5b08fc`	RBM bands to the Gazette (C4) in the return, the classification report, the provision comparison, the arrears ageing and the test; compliance PDF footer names MAIIC
`f324814` (8 October, found by the dark-mode sweep's capture of every screen)	Six ECL-over-EAD ratios divided by the string "0.00" (the sector and product-group ECL reports threw on a sector with no exposure); the report smoke now covers all 30 report methods; the licence page on a fresh database; the cache prefix keyed by environment (a permission cache filled by one database was served to another)
`742e7a7` (8 October)	The RBM Classification report put on one rule with the return (the rule itself then corrected at `f5b08fc`)

7. Test suite

The whole suite was run in one process for the first time (it had died at five minutes on every earlier attempt; see H14, fixed). Result: **577 tests, 2,779 assertions, 6 minutes 33 seconds; 455 passed, 30 skipped, 71 errors, 21 failures.**

Failing group	Count	Cause	Belongs to
Medical and inventory modules (patients, vitals, invoices, inventory purchases and sales, tariffs, SMS gateways, expense types, claim batches, appointments)	79	Models, factories and the `receptionist` role no longer exist; the tests are left over from the product the platform was forked from	The fork; retire the tests and the modules
Legacy FLI page tests (`Tests\Feature\FLI\ExternalCalculationsTest`)	5	403 on the legacy external-calculations screen and a NOT NULL constraint in its parameters table; failing on `master` before this engagement	The legacy FLI chain (M6)

Failing group	Count	Cause	Belongs to
Jetstream scaffold (`RegistrationTest`, `ProfileInformationTest`, `DeleteAccountTest`), `DatabaseTest`, `LoanApplicationApprovalStageAssignedMailTest`	8	Features disabled or models removed (`App\Models\Role` not found; `users` table absent from the hand-built sqlite schema)	The fork
Tests written in this engagement (EIR, E-Banker, staging, PD/LGD, FLI chain and route, scenario sets, compliance, Mega Farm, RBM, navigation, help)	**0 failing**		

Two observations on the suite itself. First, the engagement's tests build their own sqlite schemas by hand, so the real MySQL migrations are exercised only by the bootstrap (which does run them from nothing, so they are proven, but a column drift between a test schema and a migration would not be caught by the tests). Second, several engine tests encode the behaviour the engine has rather than the behaviour IFRS 9 requires (C3's covering test asserts the understated range; C4's test asserted the wrong bands until today): a passing test is evidence of reproduction, not of correctness, and the fixes in section 9 must change the expectations, not just the code.

8. What this means for the After Build Report's figures

As first issued, the audit said the ECL, sensitivity, stress and revenue figures of the After Build Report should not be quoted. After the closures above the clean install (`af95179`, 9 October 2026, August 2026) reads:

Figure	Before the audit	After steps 1 to 7	Basis now
ECL, undiscounted	8.94bn (12-month PD every stage; a fit the bootstrap approved; 300/401 as cash)	**9.51bn**: Stage 1 1.64bn, Stage 2 3.40bn on the lifetime PD, Stage 3 on a discounted workout LGD of 59.06 percent	IFRS 9 by stage; overlay at zero until two people approve a fit; a write-off is derecognition not cash
ECL, discounted	not run	21 loans with a locked EIR discounted through the time-phased engine; the rest carry the undiscounted figure, flagged	the time-phased engine on the governed set written on the loan
Scenario sensitivity, weighted	10.24bn beside an 8.94bn ECL	**9.88bn**; Base 9.51bn equals the allowance to the kwacha; Upside 9.06bn, Downside 10.45bn, Severe 11.51bn	the booked ECL per loan scaled by the scenario's stage PD; one weighting
Stress, base and severe	8.94bn / 11.86bn (12-month PD, DPD-only stage)	**9.51bn / 11.51bn**	the booked ECL scaled by the stressed PD and LGD, by the measured stage
Stage 3 interest (nine locked Stage 3 contracts)	on gross	on the net carrying amount: allowance 1.00bn netted, interest 17.3m, unwind 21.9m disclosed	the opening allowance the engine wrote
Revenue roll-forward, schedule-derived months	most of the Stage 3 history	3 of 483	the ledger covers a month while the feed runs past it
EIR coverage	12 of 135 loans locked	27 locked (26 with an amortisation); 20 wait only on the fee rulebook; 32 facilities now solve with their tranches at their dates	
Golden ties	9 of 9	9 of 9	unchanged by any of this

After step 8 (merged at `2903d66`) the clean install reads, at August 2026: ECL **10.00bn** (Stage 1 1.47bn on 57 loans, Stage 2 2.76bn on 25, Stage 3 5.77bn on 53), because the PD now measures defaults on the stage the

ECL is provided on (Stage 1 PD 45.0 percent, Stage 2 30.5 percent, against 35.9 and 35.2 before) and the governed three-month cure period holds six loans in their stage; sensitivity Base equals the allowance, weighted 10.38bn; the pack's gates report 3,924 of 3,924 postings in the master, 2,752 of 2,786 month-ends tying to the ledger (the 34 that do not are the one accepted-exception account), 2,134 of 2,134 month-end rows; 21 trial-balance files landed with 3,803 lines derived; golden ties 9 of 9; 140 seconds.

What still stands between these figures and a board paper is MAIIC's: the fee rulebook approval (20 EIRs, and the initial net investment of every loan with a financed fee), the moratorium type (51 schedules), D30, and the first approved run that births the golden ECL and revenue-shift numbers.

9. Order of work

Steps 1 to 7 below were done on 9 October 2026 (the commits are in section 4). Step 8 is what remains to build, step 9 the hygiene.

1 ~~Authenticate the legacy modules and bind the SQL (C7, C8)~~ done, `d58a8b0`.

2 ~~Stage 2 lifetime PD, one tenor unit, Stage 3 from the reporting date, discounted LGD (C1, C2, C3, H7)~~ done, `ba117ff`. **Status 14 October 2026:** C3 is closed; C1, C2 and H7 are partly closed (the note reads the undiscounted figure, one legacy tenor reader remains, and the LGD uses no collateral).

3 ~~Stage 3 allowance into the revenue engine; revenue after ECL; schedule fallback bounded by the feed (C5, C6)~~ done, `7d339a6`; the revenue-shift golden number is born on Dr Thom's first approved run.

4 ~~300 and 401 out of cash (H1)~~ done, `bf616a7`; the ties re-proved at 9 of 9.

5 ~~Bootstrap to the spec's rule on fits and approvals; one scenario weighting; the sensitivity from the per-scenario ECL (C9, H3, M8)~~ done, `d062a53`. **Status 14 October 2026:** H3 and M8 are partly closed (the legacy synthetic set is still seeded APPROVED; the bootstrap still locks EIRs by administrator override).

6 ~~Thresholds at the period end under maker-checker; the lock honoured by every writer; admin override governed (H11, H4, H9)~~ done, `4a31816` (thresholds are not yet proposed and approved like a setting; they are resolved at the period end with no silent default). **Status 14 October 2026:** H11, H4 and H9 are each only partly closed (see their rows), so this step is not fully done.

7 ~~Tranche drawdowns in the EIR vector; PD annualisation; post-stage splits; gross carrying in the note (H2, H8, H5, H6)~~ done, `af95179`.

8 ~~The remaining spec gaps~~ done at `2903d66` except M18 (M1, M2, M3 to M7, M10 to M14, M16; the register, the editor, the gates, the trial-balance landing, the legacy chain, the take-on recompute, cure and probation, the settings, the as-at exports). **M18** (section 16 beyond staging and PD) waits on D30. Original scope: the overlay register, the set editor, the gates at the door, the trial-balance landing, the legacy regression chain, the take-on recompute, cure and probation, the decorative settings, the as-at exports, section 16 once D30 is confirmed: two to three weeks, to be sequenced with MAIIC's decisions.

9 **Hygiene** (L1 to L6): remove `corporate/` and the ICD file, strip other-client strings, update dependencies, retire the legacy suites: a day.

Decisions MAIIC owes that the audit makes more urgent: D30 (Mega Farm basis), the fee rulebook approval (20 EIRs wait on it, as sections 5 and 8 say) (corrected 14 October 2026; it said "19 EIRs wait on it"), the default moratorium type (51 schedules wait on it), the partly-drawn interest basis (5), the prescribed RBM return form, and the first approved run from which the ECL and revenue-shift golden numbers are born.

10. Files

Audit evidence: this report; the bootstrap log and the PHPUnit log in the engagement file; the four review reports (conformance, engine, security, operations) in the engagement file; the Gazette at `docs/regulatory`. Corrections: commits `742e7a7`, `f324814`, `f5b08fc` on `eir_revenue_recognition` (master fast-forwarded).

Addendum, 14 October 2026: status of every finding against the code

Each finding was re-read against the code at HEAD `abe140d` on `eir_revenue_recognition`, plus the uncommitted working tree, which is marked as such where it matters. Since 9 October, `220a5a6` (13 October) classified and governed the write-offs and put packs 3 and 4 in the bootstrap. `216aee2` (14 October) did the following:

- kept Mega Farm out of the main PD, LGD and ECL;
- made staging and the PD read E-Banker's instalment chart in force;
- discounted the month-end ECL;
- ran the CLI steps without a time limit;
- built the tranche register, the take-on EIR check and the GL_04 fee route.

The test suite and `eir:bootstrap --verify` were not re-run for this addendum. Status words: BUILT, PARTLY, NOT BUILT, WAITING ON MAIIC.

Id	Status on 14 Oct	Evidence	What changed since 9 Oct
C1	PARTLY	Lifetime PD `ExpectedCreditLossController.php:409-421`; discounted run `FliMonthEndCommand.php:85`, `SegmentEcl.php:42`, `Bootstrap.php:347,417`; note and reports read `ecl_value` (`Ifrs9NoteService.php:192,220,488`)	`216aee2`: the month-end runs discounted, and `ecl_value` is no longer overwritten, so it holds one basis. The booked note is still undiscounted
C2	PARTLY	`InternalGradingController.php:411` reads months as years	None
C3	BUILT	`TimePhasedEclService.php`; `TimePhasedEclServiceTest`	`216aee2`: a Stage 3 loan with no recovery plan is measured at the reporting date, with no EIR needed
C4	BUILT	`RbmReturnService.php`; `RbmReturnServiceTest`	None
C5	BUILT	`EirRevenueService.php:394-407` (window = feed coverage)	None
C6	BUILT	`EirRevenueService.php:350` nets the opening allowance; the bootstrap runs the ECL every period before revenue (`Bootstrap.php:330-377`)	The ECL now runs for every period first
C7	BUILT	`RequireLogin` in `Kernel.php:45`; `RequireLoginTest`	None. APP_DEBUG must be false on the server
C8	BUILT	`TransitionProfileOptionController.php:88` casts to int	None
C9	BUILT	`Bootstrap.php` applies the route with the overlay at zero; residual: a null approver skips the check (`FliRouteService.php:71`)	None
H1	BUILT	`LoanBookBuildService.php:42-46`; `WriteOffClassifier`; `WriteOffTest` (5 tests)	`220a5a6`: GL 1064 batches and status-H closures classed as write-offs; maker-checker register `loan_write_offs`. The IFRS 7 reconciliation still shows write-offs as nil (`EclReconciliationService.php:383,416`)
H2	BUILT	`EirContractInputService.php`; tranche register `ContractInputsBuildService.php:191-260`; `TrancheScheduleTest`, `TakeonAndTrancheEirInputTest`	`216aee2`: tranches from the ledger, fee debits joined to their drawdown, schedules follow each drawdown

Id	Status on 14 Oct	Evidence	What changed since 9 Oct
H3	PARTLY	Legacy set seeded APPROVED (`Bootstrap.php:106`, `EclScenarioAssumptionSeeder`) and used for loans with no governed set (`TimePhasedEclService.php:20-25`)	The discounted path runs on the governed set on the loans (`ExpectedCreditLossController.php:531-532`, uncommitted)
H4	PARTLY	Lock checked in 11 classes; not in `MegaFarmEclService`, `EirCalculationService` or the legacy writers	`WriteOffRegisterService` checks it
H5	BUILT	Readers split by `ifrs9stage_post_qualitative`	None
H6	BUILT	`Ifrs9NoteService.php:193,220`	None
H7	PARTLY	`LgdEngineService.php:73-140`; no collateral	`216aee2`: a written-off loan left at nil is no longer counted as a cure
H8	BUILT	`PdEngineService.php:66-67`; `PdAndLgdEngineTest`	None
H9	PARTLY	Override governed (`GovernanceService.php:462`); `ScheduleWorkflowService.php:305-319` has no different-person check; `schedule_approval_control` unread	None
H10	BUILT	`GovernanceSettingsSeeder` (1900-01-01) and the backdating migration	None
H11	PARTLY	Period-end resolution in `StagingService`; no maker-checker on `staging_thresholds`; `StagingClassifier.php:84,95` uses today and 31/181	None
H12	WAITING ON MAIIC	D30. `MegaFarmEclService.php:93,166`; separation `SegmentEcl.php:69-72,121-126`, `MegaFarmProgrammeSeparationTest`	`216aee2`: programme separated, so a portfolio run no longer restores it to 100 percent
H13	PARTLY	`BaselineService.php:27-81`: 5 kinds of check against 16 rows of spec section 9	One Mega Farm line reported (not tested); method B tie scoped to in-scope GLs (`7c512ff`)
H14	PARTLY	`tests/TestCase.php:19`; `Bootstrap.php:348,495`, `SegmentEcl.php:82`, `SmokeIfrs9Reports.php:39`; 8 controllers still set 300	`216aee2`: CLI runs unlimited. ECL controller sets 300 for the web only (uncommitted)
M1	BUILT	`TrialBalanceLandingService.php` (TB_01, route TRIAL_BALANCE_FILE)	None
M2	PARTLY	`PackGates.php:144,228,273` only warn; fee GL gate `GlFeeBuildService.php:286-312` is a report	`216aee2`: fee GL gate built as a report
M3	BUILT	`OverlayService`; `OverlayServiceTest`	None
M4	BUILT	`ScenarioSetService::lock`	None
M5	BUILT	`ScenarioSetController`; `ScenarioSets.vue`	None
M6	BUILT	Legacy regression off the menu	None
M7	BUILT	`TakeonLandingService`	`216aee2`: take-on EIRs checked by `takeon_eir_plausibility` and taken through Schedule Review and the normal lock
M8	PARTLY	`Bootstrap.php:314` locks by administrator override	The run note names the override
M9	PARTLY	`ScenarioSetService.php:557`, `OverlayService.php:226`, `SeedReportFields.php:81`	None

Id	Status on 14 Oct	Evidence	What changed since 9 Oct
M10	BUILT	`PdEngineService`; `PdAndLgdEngineTest`	`216aee2`: PD cohort leaves out nil or credit starts and counts write-offs as defaults
M11	BUILT	`StagingService` (`stage_cure_months`); `StagingServiceTest`	`216aee2`: staging reads the instalment chart in force (`InstalmentChartHistory`, `InstalmentChartHistoryTest`)
M12	PARTLY	20 of 61 catalogue settings unread; `governed_parameters` remains	`partly_drawn_interest_basis`, `fee_source_precedence` and `takeon_eir_plausibility` now read
M13	BUILT	`EirAsAtController`; `EirAsAtServiceTest`	None. R12 (part-month labelled EIR) is open
M14	BUILT	`FeedZip`; `FeedZipGuardTest`	None
M15	PARTLY	`EirRevenueService.php:29` fixed; `EirLedgerCashService.php:85`; `ContractMasterImport.php:71-75`	None
M16	BUILT	`AuditFindingM16ScreensTest`	None
M17	PARTLY	`Bootstrap.php:330-361` every period; `:379-451` last period only	`216aee2`: PD, LGD, FLI route, discounted ECL and Mega Farm now run every period
M18	WAITING ON MAIIC	D30. `MegaFarmEclService.php:117-120,228-244`	`216aee2`: programme LGD on its own cohort; no typed scalar
L1	NOT BUILT	`corporate/` (715 files) and `storage/app/icd10cm_tabular_2021.xml` still tracked	None
L2	PARTLY	FDH keys in `AdjustmentMethods.php:39-40`, `TransmissionMethodCatalogue`, `GovernedValues`, `StressTestingController`; ZNBS in `TicketSeeder.php:234,253`	None
L3	NOT BUILT	No change to `composer.lock` or `package-lock.json` since 9 October (audit not re-run)	None
L4	PARTLY	Cache prefix fixed; `config/session.php:171` secure cookie unset; `VerifyCsrfToken.php:15` still excludes `file/upload`	None
L5	PARTLY	Rate above 1 read as a percent (`ScheduleWorkflowService.php:97`); `lifetime_pd` and `12m_pd` decimal(8,2); 30/360 and 30E/360 one rule (`CalculateEirService.php:87`)	`216aee2`: staging strips commas from TRANSAMT (`StagingService.php:241`)
L6	NOT BUILT	Medical, inventory and Jetstream tests remain under `tests/Feature`	None

Count on 14 October 2026:

Status	Count
BUILT	24
PARTLY	18
NOT BUILT	3
WAITING ON MAIIC	2
Total	**47**

Most consequential for 31 October 2026:

1. The FS note and reports carry the undiscounted ECL (C1).
2. The legacy synthetic scenario set is still approved (H3).
3. The controls of H4, H9, H11 and M8 are incomplete.
4. Write-offs are nil in the IFRS 7 reconciliation (H1 follow-on).
5. `--verify` checks 5 of 16 baselines (H13).
6. The hygiene items L1, L3 and L6 are not done.